

CliniScan: Milestone 3 Technical Report

Deep Learning Pipeline for Chest X-Ray Classification and Abnormality Detection

Project: CliniScan

Milestone: 3

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1. Introduction

CliniScan is a two-stage deep learning pipeline designed to automate the screening and diagnostic process for chest X-ray images. By first classifying a scan as normal or abnormal and then localizing specific findings, the system provides a comprehensive tool for radiological support. Milestone 3 details the transition from initial baselines to refined experiments involving hyperparameter optimization, data augmentation, and explainability.

2. Dataset Description: VinDr-CXR

The pipeline utilizes the **VinDr-CXR** dataset, which consists of thousands of high-resolution chest X-ray images.

- **Annotations:** Each image is annotated with bounding boxes for various abnormalities by a panel of expert radiologists.
- **Diversity:** The dataset provides a wide variety of clinical conditions, which is essential for training robust detection models like YOLOv8.

3. System Architecture

The CliniScan architecture is bifurcated into two specialized stages:

1. **Stage 1 (Classification):** Filters incoming scans. If an image is flagged as "Abnormal," it proceeds to the next stage.
2. **Stage 2 (Detection):** Uses YOLOv8s to draw precise bounding boxes around detected abnormalities, providing spatial context for the clinician.

4. Methodology & Experimental Setup

4.1 Baseline Models

Our baseline architecture comprises **ResNet18** for classification and **YOLOv8s** for detection. ResNet18 was selected for its proven performance in medical imaging, while the "small" variant of YOLOv8 offers a balance of speed and precision suitable for hospital triage systems.

4.2 Hyperparameter Tuning

We focused on the impact of the Learning Rate (LR) on the classifier's performance. Two primary rates were evaluated: **0.0003** and **0.0001**. The tuning process revealed that higher learning rates could lead to faster initial convergence but might plateau at lower final accuracies compared to more conservative schedules.

4.3 Data Augmentation

To enhance the model's robustness against clinical "noise," we employed the **Albumentations** library. Our pipeline included:

- **Spatial transforms:** Horizontal flips and rotations to account for patient positioning.
- **Pixel-level transforms:** Contrast enhancement and Gaussian noise to simulate various X-ray sensor conditions.

4.4 Transfer Learning & Alternative Backbones

We extended our experiments to **DenseNet121**. Unlike ResNet, DenseNet connects each layer to every other layer in a feed-forward fashion, which often results in better feature propagation—a critical factor when identifying subtle textures like lung infiltrates.

5. Results and Analysis

5.1 Classifier Results

The following table summarizes the performance of various classification configurations:

Model	LR	Augmentation	Backbone	Accuracy
ResNet18	0.0003	No	ResNet18	0.95
ResNet18	0.0001	No	ResNet18	<i>In Progress</i>
ResNet18	0.0003	Yes	ResNet18	0.43
DenseNet121	0.0003	No	DenseNet121	0.63

The significant performance drop with augmentation (0.43) indicates that certain medical features are likely being obscured by our current augmentation parameters, requiring a more tailored approach in future iterations.

5.2 Detector Results

YOLOv8s was evaluated on its ability to localize abnormalities within abnormal scans.

Model	Precision	Recall	mAP@50
YOLOv8s	0.55	0.37	0.41

6. Explainability and Qualitative Analysis

6.1 Grad-CAM Visualizations

To ensure the model is "looking" at the right clinical features, we implemented **Grad-CAM**. This tool generates heatmaps that highlight the regions in the X-ray that contributed most to the predicted class. Initial results show strong correlation between high-activation areas and known radiologist bounding boxes for large pathologies.

6.2 Error Analysis

A rigorous review of False Positives and False Negatives was conducted:

- **False Positives:** Often occurred in images with high anatomical variation or surgical clips.
- **False Negatives:** The model occasionally failed to detect smaller lesions or those located near the periphery of the lung field.

7. Conclusions and Future Work

Milestone 3 has provided critical insights into the strengths and limitations of the CliniScan pipeline. While baseline accuracy is high, the integration of augmentation and alternative backbones requires further refinement.

Next Steps:

- Re-tune Albumentations to use "medical-safe" transforms.
- Implement focal loss in the YOLO detector to improve recall on rare classes.
- Finalize the $LR = 0.0001$ experiments for ResNet18.